

A Pilot Search for Anomalous Mid-Infrared Excess in JWST/MIRI Archival Photometry: A Validated Pipeline

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ABSTRACT

We present a pipeline that searches JWST/MIRI archival photometry for mid-infrared excess relative to stellar photosphere predictions, developed and validated against real archive data with multiple bugs found and corrected in place. Across a randomly drawn sample of 1425 stars in 74 fields, zero excess detections survive individual vetting as confirmed candidates. Two flagged sources received detailed follow-up: one’s most distinctive spectral feature does not survive an independent model comparison and is most likely a model-comparison artifact; the other shows contamination evidence consistent with, though not yet confirmed as, a mundane explanation.

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1. INTRODUCTION

Mid-infrared excess above a star’s expected photospheric emission can indicate circumstellar dust or, in the deliberately modest framing of Dysonian SETI, waste heat from an artificial structure intercepting starlight (Wright et al. 2014). Prior systematic searches (IRAS; Carrigan 2009) (WISE; Griffith et al. 2015) were limited by their surveys’ sensitivity and resolution. JWST/MIRI offers substantially better point-source sensitivity than either at comparable wavelengths, motivating a systematic pipeline-based search of the public MIRI archive. We do not claim a completed archive survey or a detection; we present a validated anomaly-detection pipeline, its performance under real-data testing, and the individually flagged sources that followed from it.

2. PIPELINE

The pipeline follows a five-stage, tsdat-inspired architecture, each stage an explicit, traceable dataset. (1) Retriever: queries MAST for JWST/MIRI Level-3 imaging (F770W/F1000W), cross-matches against Gaia DR3 and 2MASS, and pivots per-detection rows to one row per star. (2) MIRI photometry: measures point-source flux from `i2d` mosaics via model-PSF fitting (`stpsf+photutils.psf`), not the pipeline’s own automated aperture catalog (underperforms for high-precision point-source work; Libralato et al. 2024); flux is EE-corrected to an infinite-aperture equivalent via the JWST pipeline’s own CRDS reference file. (3) Photosphere: fits T_{eff} via Kurucz (ck04models, $T_{\text{eff}} \geq 3500$ K nominal) or PHOENIX ($T_{\text{eff}} \leq 4000$ K; Husser et al. 2013), with extinction from the Bayestar19 3D dust map. (4) Excess/contaminants: flags known contaminant categories (debris disks, evolved stars, cluster membership, binarity) via literature catalogs. (5) Output: catalog, figures, tables, with every quality decision recorded as an explicit flag, not a silent cut.

3. VALIDATION

The pipeline’s rigor rests on validation against real archive data, not synthetic tests alone, and on bugs found and fixed in place rather than assumed absent. Two examples:

First, a systematic sampling bias. An early 75-field exploratory batch (74 of 75 targets succeeded and contributed stars; the remaining target is entirely coronagraphic, a separate, already-handled case, excluded from every count below) returned an impossible result across its 1425 stars — 100% single-filter detections, zero dual-band stars — prompting investigation rather than being reported as a null. The cause: a batch-capping utility took each field’s first 20 rows, but an unrelated pandas `.unstack()` operation upstream imposed an unintended ascending sort by star

ID; because Gaia-unmatched singleton detections carry small sentinel IDs and Gaia-matched stars carry ~ 19 -digit source IDs, every unmatched detection sorted before every matched star, in every field. In 71 of the 74 successfully processed fields, the cap was exhausted by unanalyzable singletons before a single usable star was ever reached — a defect confirmed to affect an earlier, smaller validation batch too. Fixed by restoring first-appearance row order and replacing the cap with random sampling; real photosphere fits rose from 1.5% to 7.9% of the sample’s 1425 stars on re-run. We report results only from this corrected, randomly drawn (not curated) 74-field/1425-star sample, not the earlier batch.

Second, a starved-fit gate. Individual vetting of this project’s first four excess-flagged sources (the same four discussed in Section 4) found two converged to near-identical, spuriously well-fit temperatures using only three mutually correlated Gaia bands (*G*, *BP*, *RP*) — effectively one color against a two-parameter model. A new check, `count_effective_bands`, treats correlated bands as one combined constraint and requires ≥ 3 independent bands before a fit is trusted, correctly separating these two failures from two well-constrained, cross-validated survivors.

A third, independent bug (Gaia/2MASS cross-match row misalignment from an unordered Python `set()` iteration) was caught pre-merge via a positional sanity check, not assumed safe. We independently re-verified both of Section 4/5’s flagged sources’ Gaia matches against this specific bug, directly: each star’s Gaia position and its real, independently-fitted MIRI PSF centroid agree to $0.23''$ and $0.10''$ respectively, both well inside the pipeline’s own $0.25''$ crossmatch radius, confirming neither match is a product of this or any other cross-match error.

4. RESULTS

Across the corrected, randomly drawn 74-field/1425-star sample, four sources initially cleared preliminary excess-significance thresholds (three dual-band, one single-band). Individual vetting — direct $\chi^2(T_{\text{eff}})$ computation, pixel-level cutout inspection, and disqualifying-flag review — resolved two as fit-quality failures caught by the starved-fit gate above. Two remain flagged after exhaustive checks (crowding, saturation, PSF-fit failure, extinction fallback, binarity, evolved-star/cluster-membership contaminant catalogs, and a dedicated pixel-level check of the local background annulus for diffuse contamination): zero confirmed candidates over the tested sample, not an archive-wide claim.

J1757132 (Gaia DR3 1633585176036702464, $T_{\text{eff}} = 3602$ K) showed a significant dual-band excess (F770W: 15.6σ ; F1000W: 4.9σ), motivating a full 8-filter MIRI SED (F560W–F2100W, $5.6\text{--}21\ \mu\text{m}$) from the archive’s own already-public multi-filter observations of this pointing (Figure 1; per-filter significance in the caption). Against the Kurucz prediction, excess at $5.6\text{--}10\ \mu\text{m}$ flips to deficit at $11.3\text{--}21\ \mu\text{m}$, crossing near $11\ \mu\text{m}$ — but this crossover does not survive an independent test: refitting with PHOENIX plus the pipeline’s own Rayleigh–Jeans extrapolation (used where PHOENIX’s $\sim 5.5\ \mu\text{m}$ -limited grid cannot reach MIRI) replaces it with a roughly flat $\sim 9\text{--}21\%$ deficit, no sign flip. The two models’ own predictions differ in shape by up to 79% relative to each other across the band (Kurucz/PHOENIX-RJ: 0.77 at F770W to 1.37 at F1500W) — enough alone to produce or erase a crossover. We conclude the crossover is most likely a model-comparison artifact, not a real spectral feature or an unexplained anomaly. The milder, uniform longward deficit that survives both models is real but far less dramatic, and degenerate with PHOENIX’s own weaker fit to this star’s real photometry ($\chi^2 = 9.9$ vs. Kurucz’s 3.2 on the same 6 bands). Two further caveats: MIRI has a documented, time-dependent chromatic flux-calibration systematic that could bias long-wavelength flux low by 3–12% if uncorrected (same direction as the deficit) — this data’s recent (Dec. 2025) calibration-reference-file version and the current calibration paper’s sub-1% post-correction per-filter uncertainty (Gordon et al. 2025) make this unlikely to dominate, though not independently re-derived here; separately, F560W/F770W’s documented “cruciform” PSF-modeling incompleteness adds $\sim 3\%$ uncertainty at exactly the two filters showing the largest apparent excess (Gordon et al. 2025).

5. AX-J1600.9-5142

AX-J1600.9-5142 (F770W: 19.1σ , $T_{\text{eff}} = 3655$ K, single-band) is a weaker case, leaning toward a mundane explanation not yet confirmed. Real, pixel-confirmed diffuse structure ($\sim 13\times$ the photon-noise level) sits in its local sky background, though a symmetric background model shows this specific structure does not measurably bias the flux measurement. The same environment argues for a related, untested mechanism instead: a cataloged dark nebula and Wolf-Rayet star sit within 1 arcmin (SIMBAD), and a bright, extended, confusing neighbor sits in WISE’s own beam — together making PSF-wing leakage the most probable explanation, though we did not perform the PSF-subtraction/deblending modeling that would confirm it.

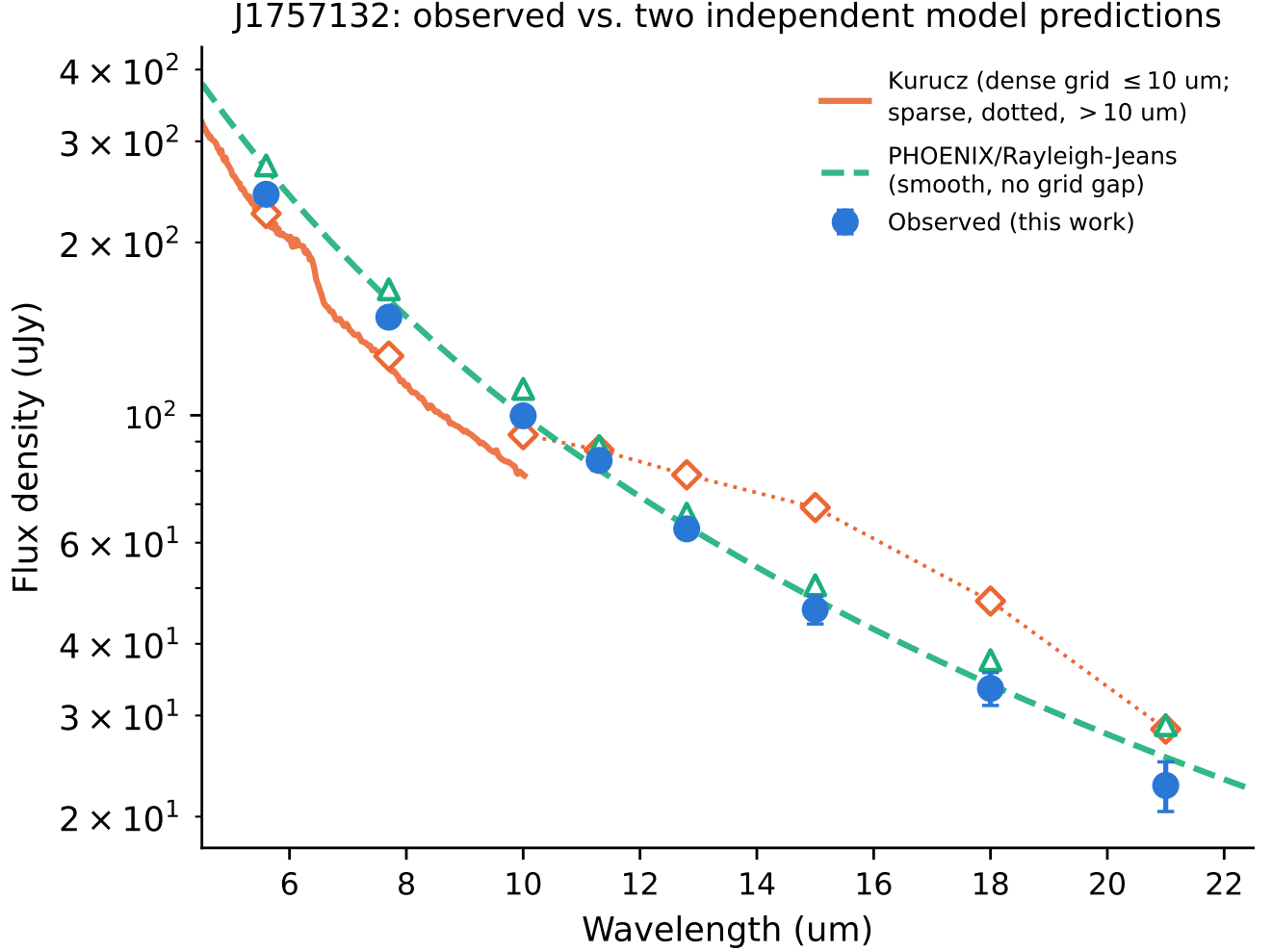


Figure 1. Observed (blue circles, this work) vs. two independent predictions for J1757132 across all 8 archived MIRI imaging filters: Kurucz (orange, $T_{\text{eff}} = 3602$ K, $A_V = 0.11$; dotted segment $> 10 \mu\text{m}$ marks a real gap in that grid’s own native wavelength sampling, Section 4) and PHOENIX/Rayleigh–Jeans (aqua, $T_{\text{eff}} = 3600$ K; smooth everywhere, no such gap). Per-filter significance relative to the Kurucz prediction: F560W $+8.0\sigma$, F770W $+15.6\sigma$, F1000W $+4.9\sigma$, F1130W -1.1σ , F1280W -7.3σ , F1500W -8.9σ , F1800W -6.3σ , F2100W -2.6σ . The two predictions visibly diverge in shape: the apparent crossover against Kurucz is largely absent against PHOENIX/Rayleigh–Jeans, direct visual evidence for Section 4’s conclusion that it is most likely a model-comparison artifact, not a real spectral feature.

6. CONCLUSION AND FUTURE WORK

This note reports a validated tool and zero confirmed candidates, not a completed search. Archive-scale application ($\sim 1000+$ stellar-classified MIRI observations) is deferred pending a candidate-triage strategy beyond simple sigma cuts, and pending two methodological gaps: validating the Rayleigh–Jeans blackbody substitute against real cool stars with known mid-IR photometry; and extending this project’s Gaia-based open-cluster membership cross-match (Cantat-Gaudin et al. 2020) to cover currently-uncovered extragalactic clusters. AX-J1600.9-5142 remains the more promising target for confirmatory PSF-subtraction follow-up.

Facility: JWST (MIRI)

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Software: astropy (Astropy Collaboration et al. 2022), astroquery, photutils, stpsf, synphot, dust_extinction, numpy

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